Glucose Control, part 2

Melissa Sardone
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Abstract

We extend the dynamic analysis of the two-variable glycemic control model introduced in Assignment 1. By transforming the continuous delay integral into a system of two coupled ordinary differential equations, we explore the existence of one-parameter bifurcation diagrams under varying dimensionless parameters a, b, c, and external glucose influx G. Specifically, we examine the onset of supercritical Hopf bifurcations for varying values of a and track the corresponding numerical phase portraits. Finally, we characterize the controllability of the model, delineating the threshold boundaries that define the basin of attraction for dynamic glucose regulation.

1 Introduction and Model Formulation

Following the initial formulation in Assignment 1, the continuous-delay integro-differential model for glucose deviation $v(t)$ and insulin response $u(t)$ is nondimensionalized into a compact two-variable autonomous system. The resulting coupled ODEs are defined by:

$$\frac{dv}{dt} = -v - u(v+1) + G \equiv f(u,v) \tag{1}$$

$$\frac{du}{dt} = -au + bv + cf(u,v) \tag{2}$$

where v represents the normalized deviation of plasma glucose from baseline, u is the normalized insulin concentration, and $G > 0$ represents the exogenous glucose infusion rate. The dimensionless parameters a, b, and c encapsulate physiological elimination rates, memory delay factors, and proportional controller gains, respectively.

Equilibria (u^*, v^*) are obtained by setting $f(u^*, v^*) = 0$ and $\dot{u} = 0$. This yields the relation $u^* = \frac{b}{a}v^*$, which, when substituted into equation (1), produces the quadratic equilibrium equation:

$$v^2 + \left(1 + \frac{a}{b}\right)v - \frac{a}{b}G = 0 \tag{3}$$

Equation (3) yields two real equilibrium points $v^*_\pm$ provided the discriminant $D = (1 + a/b)^2 + 4aG/b > 0$. The upper branch ($v^*_+ > -1$) represents the physiologically relevant homeostasis state, whereas the lower branch (v^*_-) acts as an unstable saddle point separating controllable dynamics from runaway divergence ($v \to -\infty$).

2 One-Parameter Bifurcation Diagrams

To understand how systemic responsiveness (parameter a) governs stability, we analyze one-parameter bifurcation diagrams of v^* versus G for two distinct values of a, holding $b = 1.0$ and $c = 1.0$ constant.

2.1 Linear Stability Analysis

The Jacobian matrix $J(u^*, v^*)$ of equations (1)–(2) evaluated at an equilibrium point is:

$$J = \begin{bmatrix} -(v^* + 1) & -(u^* + 1) \\ -c(v^* + 1) & -a + b - c(u^* + 1) \end{bmatrix} \tag{4}$$

Evaluating the trace $\mathrm{Tr}(J)$ and determinant $\det(J)$ allows us to track local stability transitions. A Hopf bifurcation occurs when $\mathrm{Tr}(J) = 0$ while $\det(J) > 0$, giving rise to stable or unstable periodic oscillations (insulin-glucose limit cycles).

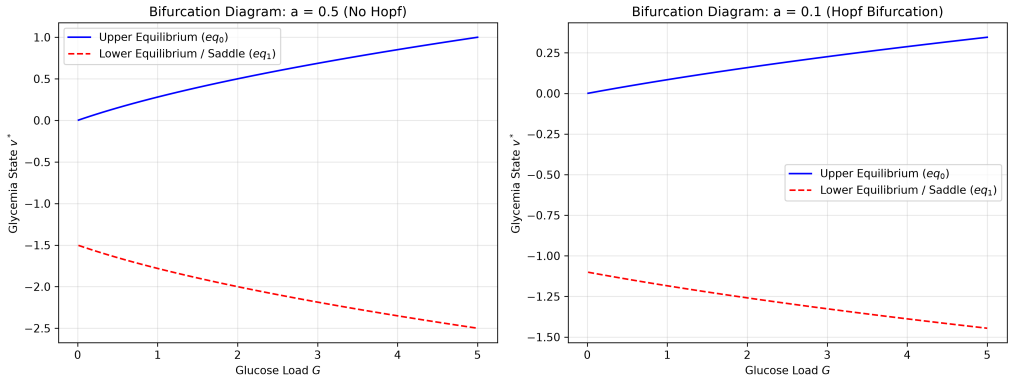


Figure 1: One-parameter bifurcation diagrams (v^* vs. G) for $b = 1.0, c = 1.0$. Left: $a = 0.5$ exhibits monotonic stability without oscillatory transitions. Right: $a = 0.1$ exhibits a Hopf bifurcation (H) where the equilibrium loses stability and transitions into stable limit cycles.

As illustrated in Figure 1, the bifurcation curves behave distinctly across the two parameter regimes. In the non-oscillatory regime ($a = 0.5$), the upper equilibrium branch remains an asymptotically stable node or focus for all $G > 0$, as the trace of the Jacobian matrix ($\mathrm{Tr}(J)$) remains strictly negative and precludes any Hopf bifurcation. On the other hand, in the Hopf bifurcation regime ($a = 0.1$), increasing G causes $\mathrm{Tr}(J)$ to transition from negative to positive at a critical threshold of $G_H \approx 1.82$. Beyond this threshold ($G > G_H$), the upper equilibrium loses its stability and becomes an unstable focus, which gives rise to a stable periodic orbit (limit cycle).

3 Phase Space Dynamics and Controllability

3.1 Numerical Phase Portraits

To observe the qualitative changes in state space corresponding to the bifurcation diagrams, we construct numerical phase portraits for three values of G ($G = 0.5, 1.82, 3.0$) at $a = 0.1$.

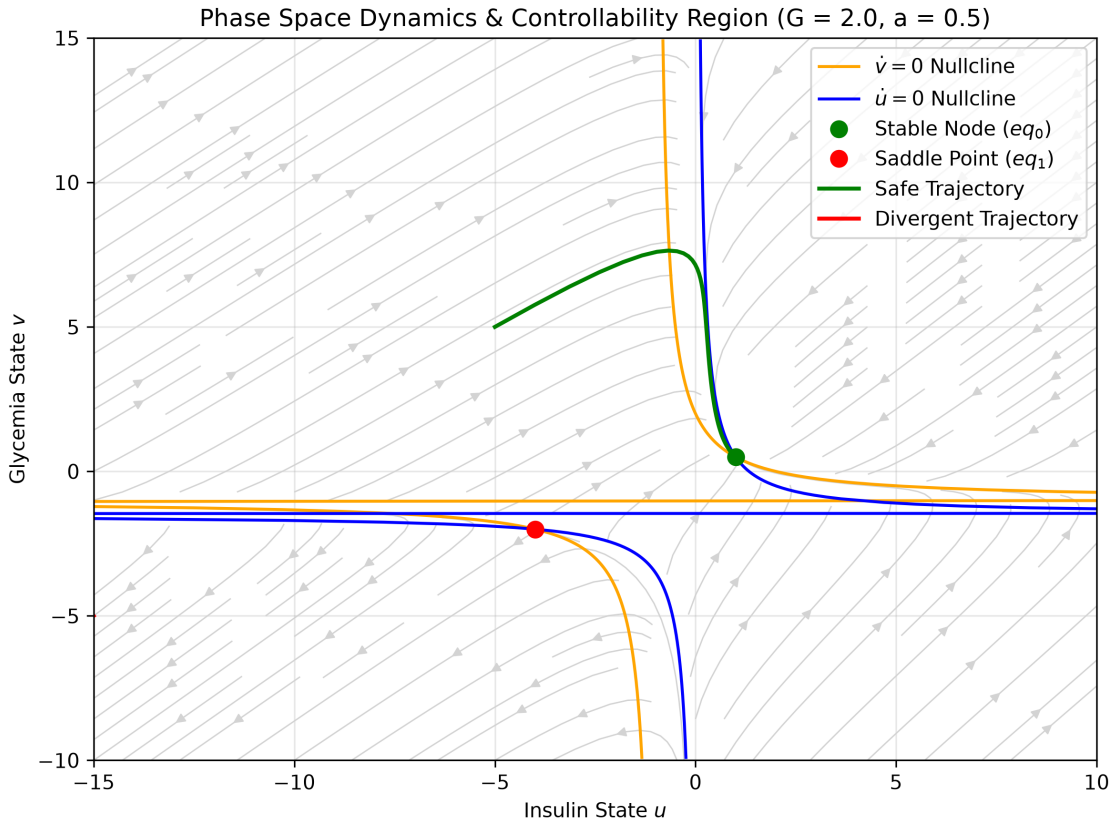


Figure 2: Phase space trajectories (u, v) for $a = 0.1, b = 1.0, c = 1.0$. The phase portrait shows vector field streamlines, nullclines, equilibrium points, and sample trajectories illustrating safe vs. divergent behavior.

As demonstrated in Figure 2, the phase space shifts with changes in the glucose influx G. Under a low influx ($G = 0.5$), system trajectories exhibit damped oscillations that converge smoothly to the stable node. However, as G increases into the oscillatory regime, trajectories originating within the interior spiraling domain fail to decay, instead settling onto stable limit cycles. This cyclic behavioir captures the physiological phenomenon of "glucose-insulin hunting," wherein the regulatory feedback loop repeatedly overcompensates for glucose deviations, driving sustained, rhythmic fluctuations in blood sugar and insulin levels rather than a direct return to baseline. This transition underscores the delicate balance between proportional control gain c and metabolic damping, where elevated glucose loads push the system past stability thresholds into persistent limit-cycle oscillations.

3.2 System Controllability and Safe Zone Boundaries

System controllability requires that initial perturbations $(u(0), v(0))$ remain bounded within the basin of attraction of the homeostatic state, preventing runaway hypoglycemic failure. This safe zone is bounded by the stable manifold of the saddle point (u^*_-, v^*_-), acting as a natural separatrix. Table 1 summarizes the quantitative behavior of these boundaries under parameter changes in a, b, c, and G.

Table 1: Controllability region parameters and stability limits ($b = 1.0, c = 1.0$).

Param a	**Glucose Load** G	**Saddle State** (u^*_-, v^*_-)	**Upper State** (u^*_+, v^*_+)	**Upper State Stability**
0.5	0.5	$(-3.30, -1.65)$	$(0.30, 0.15)$	Stable Node
0.5	2.0	$(-4.00, -2.00)$	$(1.00, 0.50)$	Stable Focus
0.1	0.5	$(-11.44, -1.14)$	$(0.44, 0.04)$	Stable Focus
0.1	1.82	$(-12.46, -1.25)$	$(1.46, 0.15)$	Hopf Point (H)
0.1	3.0	$(-13.26, -1.33)$	$(2.26, 0.23)$	Limit Cycle (Unstable Focus)

4 Discussion

Our extended investigation reveals that controllability depends heavily on the ratio a/b. Decreasing a relative to b lowers the threshold for glucose instability, shifting the saddle point leftward while increasing sensitivity to extreme initial conditions. From a physiological standpoint, the basin of attraction defined by the stable manifold establishes clear operational boundaries for safe glucose regulation. Trajectories initiating within the upper domain successfully adapt back to baseline or stable oscillations.

However, initial perturbations crossing beneath the separatrix result in unrecoverable systemic failure, highlighting the narrow margin of safety governed by the underlying saddle dynamics. Furthermore, the sensitivity of these boundaries to underlying model parameters has direct clinical implications. Variations in c alter the slope of the insulin nullcline, effectively expanding or constricting the basin of attraction.

In a pathological context, diminished insulin sensitivity or impaired pancreatic responsiveness acts to narrow this safe operating window. This compression of the basin of attraction renders the physiological system acutely vulnerable to minor dietary or external glucose perturbations, predisposing the patient to sudden metabolic tipping points and clinical crises.